

FARHAN MUJEEB

Embedded Systems & Firmware Engineer

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with hands on experience in embedded firmware development, microcontroller programming, sensor interfacing, BLE communication, and IoT systems. Skilled in Embedded C, Python, STM32, Nordic nRF54L15, Zephyr RTOS, nRF Connect SDK, UART, SPI, I2C, and Modbus RTU. Built projects involving BLE asset tracking, RSSI-based proximity estimation, wireless backhaul, autonomous robotics, and remote monitoring with automated alerts. Seeking an entry-level Embedded Firmware, Embedded Systems, IoT, or Wireless Product Development role.

CORE TECHNICAL SKILLS

Programming Languages: C, Embedded C, Python

Microcontrollers: STM32 (F072B), Nordic nRF54L15, ESP32

Embedded Systems: Firmware Development, GPIO, Interrupt Handling, Sensor Interfacing, EEPROM, Hardware Debugging, CubeMX

Communication Protocols: UART, SPI, I2C, Modbus RTU

Wireless & IoT: Bluetooth Low Energy (BLE), Wi-Fi, LoRa, MQTT, RSSI

RTOS & SDKs: Zephyr RTOS, nRF Connect SDK, Arduino IDE

Development Tools: STM32CubeIDE, VS Code, Logic Analyzer, LTspice, KiCad

Hardware: PCB Assembly, PCB Inspection, Flame Sensors, Gas Sensors, Temperature Sensors, Motor and Pump Control

PROJECTS

Indoor BLE Asset Tracking System

- Developing a BLE-based indoor asset-tracking system using Nordic nRF54L15 anchor nodes and configurable BLE tags.
- Implemented BLE scanning, tag detection, and RSSI measurement for proximity estimation.
- Designed a wireless backhaul architecture using Wi-Fi and MQTT for cloud data transfer.
- Integrated BLE communication, wireless data transmission, and embedded firmware components for indoor asset monitoring.

Fire-Fighting Robot

- Designed and developed an autonomous fire-fighting robot using flame and gas sensors for fire detection.
- Implemented sensor-based navigation and automatic movement toward the detected fire source.
- Integrated a water pump and motor-control system for fire extinguishing.
- Implemented SMS and phone-call notifications to alert registered users during fire incidents.
- Tested sensor response, motor movement, pump activation, and alert functionality under different operating conditions.

Smart Storage Cabinet

- Developed a prototype automated medication storage and monitoring system.
- Integrated temperature sensors, dual-chamber temperature control, image-recognition concepts, and embedded hardware.
- Implemented remote monitoring and automated alert functionality.
- Contributed to hardware integration, software development, and system-level testing.

INTERNSHIPS & TRAINING

Technomake — PCB Assembly Intern

Jun 2025

- Gained practical exposure to PCB assembly, component placement, soldering, and manufacturing workflows.
- Performed visual inspection of assembled PCBs to identify component-placement and assembly defects.
- Reviewed PCB assemblies for visible quality issues and manufacturing inconsistencies.

IPCS — Machine Learning Trainee

Jun 2025

- Completed technical training in machine learning concepts and practical applications.
- Studied fundamental machine-learning workflows, model concepts, and application areas.

Tech-By-Heart — Machine Learning Trainee

May 2023

- Attended technical training on Machine Learning fundamentals and practical applications.

EDUCATION

B.Tech – Electronics and Communication Engineering | Albertian Institute of Science and Technology, Kalamassery | APJ Abdul Kalam Technological University | 2022–2026

CGPA: 8.01 / 10

Higher Secondary (CBSE): 80% | Gregorian Public School | 2021–2022

Secondary (CBSE): 88.8% | Gregorian Public School | 2019–2020

CERTIFICATION

Machine Learning and IoT — Reverttech IT Solutions (Jan 2025 – Feb 2025)

LANGUAGES

English, Malayalam, Hindi